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Optimized Implementation of Gift Cipher

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Abstract
The Internet of Things is an emerging area which deals with transfer of the data through the wired or wireless network. The prime thing that needs to be addressed in this is the security of the data that must be transferred within the optimized time limit. In this paper, throughput and time delay are need to be considered for the optimized data transfer and while concentrating on this, there is a possibility of allowing the data to be vulnerable to attacks. Security algorithms currently available may be adequate for the wired system and not as the same for wireless scenario. PRESENT cipher is a one of the popular cryptosystem used in wireless which falls under the light weight cryptography category. Gift cipher is an enhanced version of PRESENT cipher. Which aims that maximizing the throughput. In this, iteration structure used for encryption. This can still be improved and optimized in terms of increased data rate and reduced time delay. In this paper, implements the optimization technique of the existing GIFT cipher and throughput is considered as the performance metrics. Pipeline and sub-stage pipeline techniques are used for enhancing the architecture.

Keywords Light weight cryptography · Gift cipher · Iterative · Pipelining · Sub-stage pipelining

1 Introduction

The IoT, introduces new kind of network environment in which not only the devices but also things are connected and shares data. Here thing refers to anything which is smart in nature and possess some data. Objects like keychain, pillow and handbag which can be anything is made smart and can share data. The network consisting of these kind of things may be small in size and having restriction in usage of battery power. Any operation like data transfer from/to the device should be done by using negligible amount of power. In this background, the safety of the data entirely relies on cryptographic algorithm

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embedded in the device. Without the algorithm confidentiality and integrity of the data it will be highly questionable. The confidentiality is affected the genuiness of the source and destination cannot be ensured. If the integrity is compromised, then the data may get tampered or altered in the transition. To thoroughly avoid these kind of consequences, a perfect cipher must be embedded in the smart device. Optimization must be done in order to fit the cipher in the smart environment. This paper discusses about the optimization technique carried over GIFT cipher [1, 14] in order to make fit in to IoT [11]. For this pipelining and sub-stage pipeline are used and the paper is organized as follows. Section 2 following this introduction discusses the existing iteration model of GIFT cipher, followed by the overview of the GIFT cipher in the section 3 and section 4 discusses about the proposed work followed by the result and analysis in section 5. Conclusion and future work followed by the reference.

2 Existing Optimization Techniques

The main objectives of the optimization technique are increasing the throughput, decreasing the area, reducing the latency, and improving the power efficiency. Some of the existing optimization techniques on advanced encryption standard (AES) are presented as literature survey in the following sub sections. The prime reason for choosing AES is that, it is similar to Gift cipher in the substitution–permutation (SP) box.

2.1 Folded Register and Parallel Optimization Technique

Rahimunnisa et al. [10] implemented the folded register with a parallel operation. In this procedure, each word was operated parallel. The waiting time was dramatically reduced and thus resulted in high reduction of latency. Chodowiec and Gaj [4] implemented the folded register optimization technique on AES. The AES operation was performed over the sub unit called word and these words underwent the process of step by step as per the flow of the algorithm. The speed was reduced in the folded register operation. The throughput of folded with parallel optimization (37.1 Gb/s) was more than folded register optimization (1.3 Gb/s) [3, 9].

2.2 Inter and Intra Pipelined Optimization

Yoo et al. [13] implemented the inter and intra pipelined and parallel structure of AES algorithm which resulted in improved performance at the cost of increased complexity. Due to the parallel architecture the throughput was high 29.77 Gb/s. Internal pipeline structure would reduce the clock cycle. Hodjat and Verbauwhede [7] implemented the pipeline concept, only for inner and outer rounds. It was different from the optimization of Yoo et al., since it entirely pipelined. They achieved a throughput of 4.2 Mbps/Slice. but low resource was utilized [6, 13].

2.3 Partial and Dynamic Reconfigurable Architecture

Ismaili and Moussa [12] tried the optimization technique on AES, but the implementation was carried out without pipelined and parallel architecture. Criado et al. [6] proposed

the partial and the reconfiguration of AES on the FPGA platform [8]. In this parallel and pipelined implementation of architecture reduced the delay and increased the performance of the cipher. The throughput was high 24.922 Gb/s. So, partially power was utilized but throughput was lower than the Criado et al. implementation. In this case, the throughput was 1.6 Gb/s [6, 12].

On the above literature review shows that pipelined and parallel architecture will increase the throughput. Sub-stage pipeline will increase the performance of the SP box block cipher. The reconfigurable architecture will reduce the hardware utilization. Overall depicts the tradeoff between throughput, hardware utilization and performance of the cipher. So the pipelined and sub pipelined concept were adapted for recent light weight gift cipher.

3 Gift Cipher Overview

Gift is round based block cipher proposed by Banik et al. [2] with the advancement of Present cipher. The main purpose of this cipher was to protect from the hull attack. This complexity of the Gift cipher was increased though the non-linear function of S-box. It was strong towards the differential attack and provided better performance. The implementation on FPGA is simple and straight forward method. The round-based function was followed for the cipher. The utilization of power was low when compared with the other substitution and permutation block cipher. The Sub cells (S-box) complexity decides the strength of light weight algorithm. Here, the Present cipher is vulnerable to linear hull attack at 20th round due to S-box simplicity [5].

The Gift cipher consists of three steps: Sub cells, Perm-bits and add round key. It performs in variable block size and key size.The input 64-bit requires 28 rounds to encrypt the data. The input 128-bit requires 40 rounds to encrypt the data. For both, key size is same and it requires 128-bit. In the process of initialization, the constant value is initialized with plaintext in the form of nibbles that means groups of 4-bit. The key is split into bytes (groups of 8-bit). Each round function in the round_cipher_data includes Sub cells followed by Perm-bits. At last, Round Key is XORed with predefined constants. The S-Layer/Sub-cells include the S-box where each nibble of plain text is assigned to the S-box.

The Perm-bit operation permutes the Sub-cells output between the position k and P(k). The add-round-key function involves 64-bit and 128-bit key with 7-bit round constant as part of its state operation. From each state (counter size is 6 bits), 128-bit round key is generated. The key is split into 8-bit words and at last two words are extracted. Then the shifted data are appended at the front and the round constant is obtained from the linear feedback shift register (LFSR). The input data in state registers are processed through a series of round functions. In each round, S-layer, bit wiring layer and add-round-key implementations are executed with previous round cipher state. After the 40 rounds of 64-bit data, the current state will be provided as the final encrypted output [1, 2, 4, 7].

3.1 The Iterative Architecture of Gift Cipher

The implementation using the iterative technique is a straight forward method. The advantage of this method is low hardware requirement. The 64-bit input is processed through the multiplexer. In this 2×1 multiplexer is used. The first line is used to process the input. The second line will process the output from the add-round key. In this, the number of rounds is

equal to the number of clock cycles. The data from the multiplexer is temporarily stored in state register. The 64-bit input message from state register refers the Sub cell (S-box). Each four bit represents the one hexa decimal number in the substitution box which is similar to the Present cipher [2] as the S-box is based on some non-linear functionality, it can be seldom attacked by linear or differential attack procedures. The output from the s-box is re-indexed or bit permutated in a P-box layer (Fig. 1).

Thus the Present cipher and Gift cipher differs in the first and the third stages. The add-round key and bit wiring are not straight forward like a Present cipher. In this, 64-bit follows certain procedures. The representation of bits is as follows for message (PT), output of Add round key, Key (RK), Cipher text (CT) and state counter (CS):

The plain text 0th bit (PT_0) is XORed with round key (RK_0) to produce cipher text (CT_0). CT_1 also is generated in the same way. CT_2 will be referred by PT_2. CT_3 and CT_4 follow XORing procedure. CT_5 is XORed with counter value. The scenario repeats as groups for all the 64-bit. The iterative structure preferred for low area application comparatively yields less throughput than that of pipelined and sub-stage pipelined architecture [1, 9].

4 Proposed Work

4.1 Pipeline Optimization of Gift Cipher

The process of pipelined architecture is similar to the iterative structure with some added register either at the beginning of the round or at the end of the round (also known as round level pipelining) leading to high level throughput (Fig. 2).

The simultaneous process of all the 16 blocks of the input increases hardware utility compared with the iterative structure. Still that utility is less when compared with that of the sub-stage pipeline (or) unrolled model. In this structure the speed may be increased further by replacing the register with retiming register (with additional complexity). The pipeline structure is preferred for high throughput and performance.

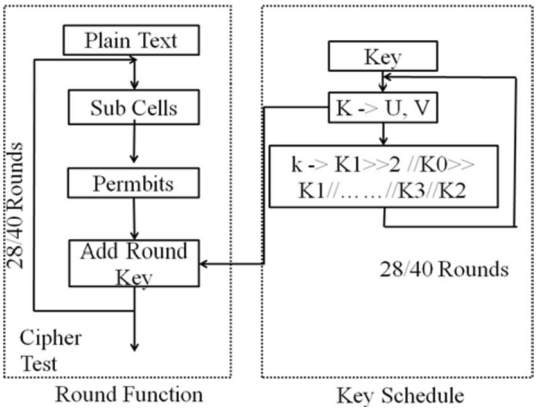


Fig.1 Iterative structure of gift cipher

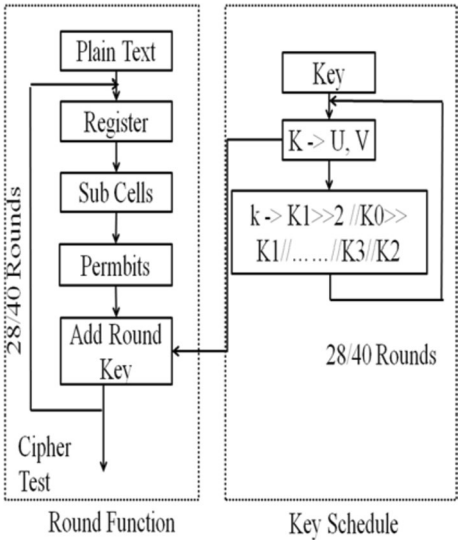


Fig.2 Pipelined architecture of gift cipher

4.2 Sub-stage Pipeline Optimization of Gift Cipher

The sub stage pipelining buffers the data path between each step of algorithm. The row of register is added between plain text and substitution box (sub cells). The second row of registers is added between substitution box and permutation box (Perm bits).The third row of register data is added between perm bits and add round key. Then, the key

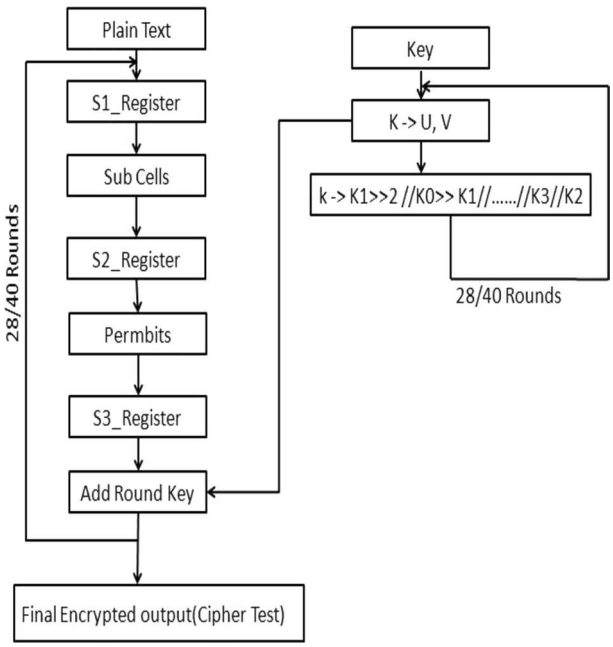


Fig.3 Sub-stage pipelined structure of gift cipher

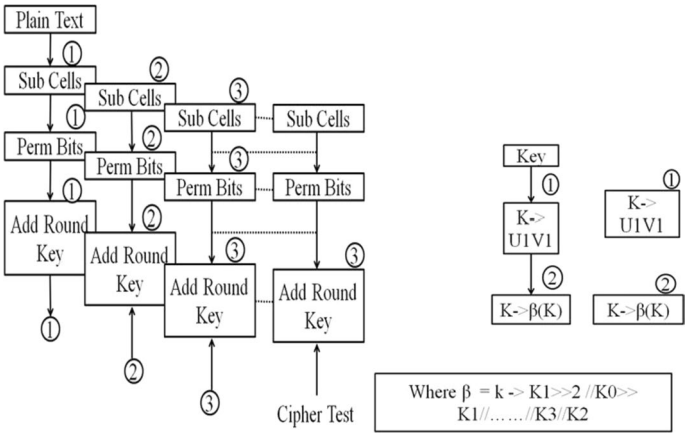


Fig.4 Loop Unrolled structure of gift cipher

Table 1 Comparison of iterative, pipelining, and sub-pipeline for gift cipher

Gift architecture	Plain text bits	LUTs	CLB slices	Minimum clock period (ns)	Maximum frequency (MHz)	Throughput (Gbps)
Iterative	64	78	20	2.01	497.51	31.84
Pipeline	64	1792	632	1.75	571.43	36.57
Sub-pipeline	64	2813	1080	1.71	584.8	37.43

Plaintext: 64 bit Key:128 Round:28

data is obtained for each clock. The key scheduling process is completed before the round function is completed. It provides higher throughput. The performance is more than that of Iterative and pipeline architecture (Fig. 3).

4.3 Loop Unrolled Model of Gift Cipher

The unrolled architecture provides output in a single clock. but it consumes large amount of resource. If the application demands time boundness as high priority, then the implementation of the unrolled architecture is justified. Otherwise, keeping the very huge complex hardware requirements as the rejecting factor, the architecture may be excluded from the group of optimization structures considered for comparison (Fig. 4).

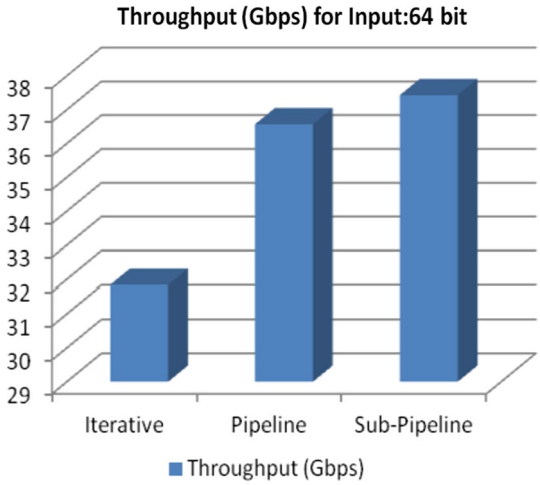


Fig.5 Throughput achieved in various models of gift cipher for the plain text is 64-bit

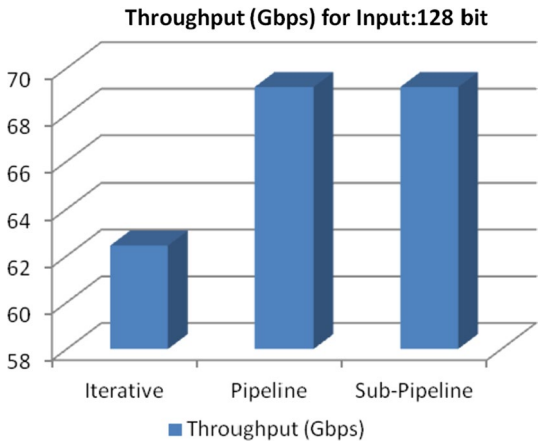


Fig.6 Throughput achieved in various models of gift cipher for the plain text is 128 bit

5 Results and Discussion of Gift Cipher

As the proposed system is the improved version of Gift cipher with conventional iterative structure with the results obtained though the pipelined and sub-pipelined architecture are compared with that of the iterative type and Table 1 depicts the comparison.

Existing Gift architecture is a resource constrained one and hence having limited throughput. To increase the throughput an array of register is added which forms the pipeline architecture. The increase in number of register leads to increase in LUT (Look up Table) which in-turn result in increased combinational logic blocks (CLB).The effect of this overhead will not be held much when compared to the gain the obtain in terms of throughput.

The proposed structure which improves the throughput, when compared to an iteration architecture is graphically represented in Fig. 5.The pipelined architecture shows

Table.2 Comparison of iterative, pipelining, and sub-pipeline for gift cipher

Gift architecture	Plain text bits	LUTs	CLB slices	Minimum clock period (ns)	Maximum frequency (MHz)	Throughput (Gbps)
Iterative	128	144	36	2.05	487.8	62.44
Pipeline	128	5120	1280	1.85	540.54	69.19
Sub-Pipeline	128	7857	2536	1.85	540.54	69.19

Plain Text: 128 bit Key:128 Round:40

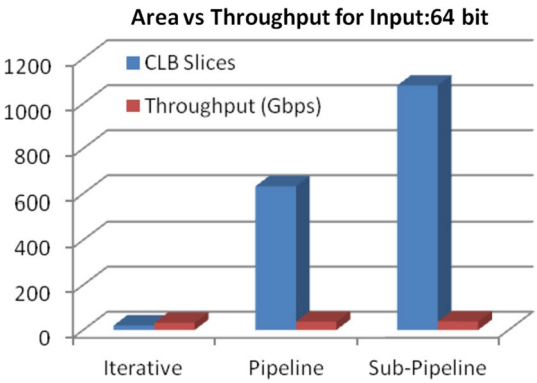


Fig.7 Area versus throughput for the plain text is 64 bit

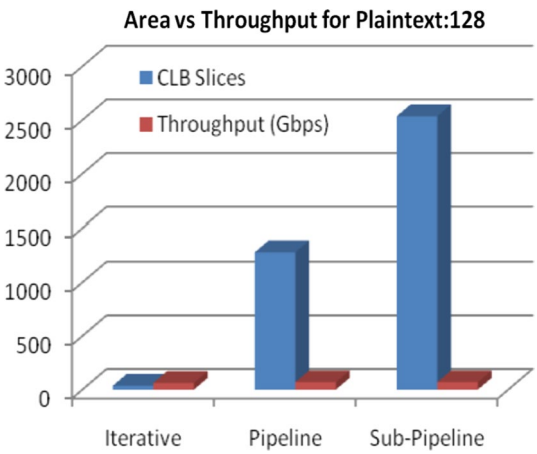


Fig.8 Area versus throughput for the plain text is 128-bit

the improvement of 13% and sub pipelined structure render 15% more in terms of throughput. This is for 64-bit system and the same is done for 128-bit system and the result is portrayed in Fig. 6.

In this simulation, even though there is an improvement of 9.75% in the throughput for the pipelined architecture, the magic can not be extended by adapting sub-pipelined structure (which is also rendering the same result).The related experimental data for this is given in Table 2.

Table.3 Test vector I

Input: 64 bit Output: 64 bit Key: 128 bit Round: 28	
Plain text	06034F957724D19D
Cipher text	E8D3E0B8C989F15B
Key	9EB93640D-088DA6376A39D-1C8BEA71E1

Table 4 Test vector II

Input: 128bit Output: 128 bit Key: 128 bit Round: 40	
Plain text	F20ADB0EB08B648A3B2EEED1F0ADDA14
Cipher text	F3D280D6464A6F16AE09683FE4AE655F
Key	4F55CFB0520CAC52FD92C15F37073E93

The comparison provided for Iterative, Pipeline, and sub pipelining optimization techniques on gift cipher is shown in Tables 1 and 2. The graphical Analysis is presented in Figs. 5, 6 and 7. The iterative architecture has utilized less resource. Pipelined and sub-pipelined techniques have achieved the high throughput. It demonstrates the trade-off between area utilization and throughput and is represented in Fig. 7.

The hardware complexity for the 64-bit system is illustrated in Fig. 7 and the same for 128 system in Fig. 8.

The synthesis has been carried out using Xilinx tool with the following specifications:

Synthesis Tool: Precision RTL Synthesis 64-bit 2015.2.10
IDE: Xilinx ISE 14.7
Xilinx FPGA Device: 7VX690TFFG1930

The resulting test vector data is shown in Tables 3 and 4 for 64 bit and 128 bit respectively.

6 Conclusion

The optimization of the iterative Gift cipher is done by constructing pipelined and sub-pipelined architecture at the cost of increased system complexity. The resulting optimized structure is more suited for smart devices which incorporates more memory and limited in battery usage. Depending upon the memory size, the type of optimization can be chosen. Say, without optimization, the conventional system is suited for passive RFIDs, the pipelined optimization may be used in semiactive RFID tags and the sub-pipelined structure is suitable for active RF devices (web camera, floor control system), etc. The selection of optimization technique and its implementation is always a trade-off depending upon the requirement.

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